UNIVERSITÉ DE FRANCHE-COMTÉ

CRESE
CENTRE DE RECHERCHE
SUR LES STRATÉGIES ÉCONOMIQUES

Panel Data Econometrics

Chapter 2 – The Error Component Model

… and some estimators

Reminder

If we focus on the individual level, the total variability of a measure can be decomposed as the sum :

intra-individual variability + inter-individual variability

In the previous chapter, we introduced the between and the within operator.

- Between operator (B): transforms all values into the individual mean. It removes all intra-individual variability.

- Within operator (W): transforms all values into distances to the individual average. It removes all inter-individual variability.

Introduction

In this second chapter, we discuss …

Part 1 – Notations and hypotheses

Part 2 – Introduction to the application

Part 3 – The pooling model

Part 4 – The between model

Part 5 – The within model

Part 6 – Comparison of the estimators

Notations and hypotheses

The error component model

$$y_{it} = \boldsymbol{x}_{it}\boldsymbol{\beta} + \varepsilon_{it}$$

We now consider that the error is the sum of two effects :

- η_i is the individual effect for individual i.
- ν_{nt} is the idiosyncratic (residual) effect.

Therefore:

$$\varepsilon_{it} = \eta_i + \nu_{it}$$

Notations and hypotheses

Hypotheses regarding the errors

The unconditional expectation of η is 0 :

$$E[\eta_i] = 0$$

η is homoskedastic and mutually uncorrelated

$$E\left[\eta_i^2\right] = \sigma_\eta^2 \quad ; \quad E\left[\eta_i \eta_j\right] = 0$$

For now, we also assume :

η is exogenous, in that it is not related to variables in $\boldsymbol{X}$

$$E[\eta \mid X] = 0$$

Notations and hypotheses

Hypotheses regarding the errors

The unconditional expectation of ν is 0 :

$$E[\nu_{it}] = 0$$

ν is homoskedastic and uncorrelated :

$$E\left[\nu_{it}^2\right] = \sigma_\nu^2 \quad ; \quad E\left[\nu_{it}\nu_{ju}\right] = 0$$

ν is exogenous, in that it is not related to variables in $\boldsymbol{X}$:

$$E[\nu \mid X] = 0$$

Notations and hypotheses

Hypotheses regarding the errors

The two components of the error are uncorrelated :

$$E[\eta_i \nu_{it}] = 0$$

The variance of one error is

$$E\left[\varepsilon_{it}^2\right] = \sigma_\eta^2 + \sigma_\nu^2$$

The covariance between two errors from the same individual :

$$E[\varepsilon_{it}\varepsilon_{iu}] = \sigma_\eta^2$$

The covariance between two errors from different individuals :

$$E\left[\varepsilon_{it}\varepsilon_{ju}\right] = 0$$

Notations and hypotheses

Hypotheses regarding the errors

For a given individual, the variance-covariance matrix of the error is given by :

$$\Omega_i = E[\varepsilon_i \varepsilon_i'] = \sigma_\nu^2 I_T + \sigma_\eta^2 J_T$$

For the full sample, the variance-covariance matrix of the error is a block diagonal matrix given by :

$$\Omega = I_N \otimes \left(\sigma_\nu^2 I_T + \sigma_\eta^2 J_T\right) = \sigma_\nu^2 I_{NT} + \sigma_\eta^2 (I_N \otimes J_T)$$

Notations and hypotheses

Hypotheses regarding the errors

$$\Omega = I_N \otimes \left(\sigma_\nu^2 I_T + \sigma_\eta^2 J_T\right) = \sigma_\nu^2 I_{NT} + \sigma_\eta^2 (I_N \otimes J_T)$$

Remembering $B = \frac{1}{T} I_N \otimes J_T$ and $W = I - B$, we can decompose the variance of the error in within and between components :

$$\Omega = \sigma_\nu^2 (B + W) + T\sigma_\eta^2 B$$

With $\sigma_l^2 = \sigma_\nu^2 + T\sigma_\eta^2$

$$\Omega = \sigma_\nu^2 W + \sigma_l^2 B$$

About the illustrations

The research question

Schaller, H. (1990). A re-examination of the q theory of investment using us firm data. *Journal of applied econometrics*, *5*(4), 309-325.

We are interested in the relationship between Tobin's Q (measure of over/under-valuation of a firm) and investment in a firm.

About the illustrations

The data

We get the data from the pder package.

It includes, among other variables :

cusip : compustat's identifying number

year : year

ikn : investment divided by capital : narrow definition

qn : Tobin's Q : narrow definition

The Pooling model

Let's ignore the panel dimension, and consider the *raw data*. This is what we call the pooling model.

We perform an OLS estimation, as we did in the previous econometrics lectures.

$$\widehat{\boldsymbol{\beta}}^{OLS} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

What are the properties of the OLS estimator ?

The Pooling model

If all aforementioned hypotheses hold:

$$E\left[\widehat{\boldsymbol{\beta}}^{OLS}\right] = \boldsymbol{\beta}$$

$$\boldsymbol{var}\left[\widehat{\boldsymbol{\beta}}^{OLS}\right] = ?$$

$$\boldsymbol{var}\left[\widehat{\boldsymbol{\beta}}^{OLS}\right] = (\boldsymbol{X}'\boldsymbol{X})^{-\mathbf{1}}\boldsymbol{X}'\boldsymbol{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-\mathbf{1}}$$

But $E[\varepsilon\varepsilon']$ is not $\sigma^2\boldsymbol{I}_{\boldsymbol{NT}}$ anymore ! It is $\boldsymbol{\Omega}$! We find ourselves in a situation very similar than the one induced when hypothesis A4 is not respected.

The Pooling model

To summarize :

In the error-component model, the OLS estimator is …

- Unbiased
- Consistent

but …

- Its variance-covariance matrix is not $\sigma^2(X'X)^{-1}$, we have to correct it to interpret the coefficients.
- The OLS estimator is not efficient. There exists an estimator (relying on GLS) that is also unbiased and provides a better accuracy.

The Pooling model

Application

- See R code –

- We estimate the pooled estimator with OLS.
- We estimate the pooled estimator with plm.
- We discuss about the correction of the standard-errors.

The Between estimator

The between estimator is a model in individual mean. We transfom all variables in individual mean, and we use the OLS estimator.

$$By = BX\beta + B\varepsilon$$

We therefore remove all notion of intra-individual variability, and come close to a cross-sectional model.

Note: some variables will be unchanged by the transformation!

The Between estimator

The between estimator can therefore be written as :

$$\widehat{\beta}^{B} = (X'BX)^{-1}X'By$$

The variance of this estimator is expressed as :

$$V\left(\widehat{\beta}^{B}\middle|X\right) = \sigma_l^2(X'BX)^{-1}$$

and σ_l^2 can be estimated as $\widehat{\sigma_l^2} = \frac{e'e}{N-K}$

The between estimator is consistent, and asymptotically normal.

The Between estimator

Application

- See R code –

- We estimate the between estimator « manually »
- We see that the standard error (and the degrees of freedom) are not correct if we keep all repeated observations in By and BX
- We estimate the between estimator with plm

The Within estimator

The within estimator is a model in deviation to the individual mean. We transfom all variables in deviation individual mean, and we use the OLS estimator.

$$Wy = WX\beta + W\varepsilon$$

We therefore remove all notion of inter-individual variability.

Note: some variables will just drop from the estimation!

The Within estimator

The within estimator can therefore be written as :

$$\widehat{\beta}^{W} = (X'WX)^{-1}X'Wy$$

The variance of this estimator is expressed as :

$$V\left(\widehat{\beta}^{W}\middle|X\right) = \sigma_\nu^2(X'WX)^{-1}$$

and σ_ν^2 can be estimated as $\widehat{\sigma_\nu^2} = \frac{e'e}{NT-N-K}$

The within estimator is consistent, asymptotically normal, and **asymptotically efficient**.

The Within estimator

Application

- See R code –

- We estimate the within estimator « manually »
- We see that the standard error are not correct (and the degree of freedom is too high)
- We estimate the between estimator with plm

Comparison of the estimators

The between and the within estimators focus on only one source of variance, respectively:

- The inter-individual variability
- The intra-individual variability

The pooled estimator uses both sources of variance. We can expect that the coefficients estimated will lie somewhere between the within and the between estimator.

Important: if there is no correlation between the errors (ε) and the covariates (X), then all estimators are equivalent in large samples!

Comparison of the estimators

The OLS estimator is actually a weighted average of the within and the between estimators.

The weights are basically related to the part of intra-individual variability and the part of inter-individual variability in X.

$$\hat{\beta}^{OLS} = (X'WX + X'BX)^{-1}(X'WX\hat{\beta}^{W} + X'BX\hat{\beta}^{B})$$

$(X'WX + X'BX)^{-1}(X'WX)$ is the percentage of the total variance in X that can be explained by **intra-individual** variance.

$(X'WX + X'BX)^{-1}(X'BX)$ is the percentage of the total variance in X that can be explained by **inter-individual** variance.